

Report – FIFO Paging Simulation

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Course: COMP2240 – Operating Systems

Assignment: A3 – Paging Simulation

1. Testing Approach and Methodology

I felt testing crucial and rather difficult part of this assignment due to the nature of multiple file input, by utilising the given sample files I was able to test my program to achieve the right outputs. I employed several strategies to help me test my program both during construction and after.

- To save time when running the program, I created a piece of code which would just take the entire sample folder as the input. Since most of my issues was in small numerical differences in the turnaround time or the fault times, this allowed me to make quick changes and test immediately.
- I kept the frames and CPU quantum values consistent throughout development to help get consistent results so I can pinpoint issues.
- After completing the program, I tested multiple different values for the quantum and changed around the contents of the process files to ensure there were no issues.

2. Comparison of Resident Set Management Policies

• Fixed-Local Replacement:

During my testing I was able to discover that this method offered a stable and consistent way to process pages. It seems that it is very dependent on the number of frames allocated compared to number of pages/processes it must do if number of memory frames are sufficient it can produce a great result with minimal faults. However, if they aren't it can lead to high turnover and high faults such as in sample 2, especially when comparing to variable globe replacement.

• Variable-Global Replacement:

The flexible nature of this method can offer great benefits in some cases and is negligible in others. For example, in sample 1 it offers to benefits compared to fixed-local, whereas in sample 2 the benefits are amazing, less faults and quicker turnover. Form testing it seems that for larger data sets it seems to be consistently quicker than the fixed-local. But there's a catch the frames need to quite a lot higher for this to be case, for example when I was using 10 frames fixed local was achieving far better results than variable global.

So to conclude based on my own observations and the sample results, both methods have their own uses cases, variable is great if its larger data set and

you have substantial memory frames for it, but for smaller sets fixed-local works fine and often using lower frames.

Observation: For Sample2, Variable-Global with 10 frames resulted in slightly higher page faults and turnaround than expected. Increasing frames to 20 aligned the results with assignment expectations.

3. Edge Cases Considered

- **Processes requesting only a single page** – ensured page fault handling still works correctly, this was already a requirement to achieve the results of sample 1.
- **Repeated Page Requests** – if a process requires same page multiple times, I had to ensure that the page wouldn't keep reloading and creating a bunch of faults.
- **Small frame allocation** – Ensured program would still run when there's more pages than frames, even if not effectively
- **Small frame allocation in fixed local** – Using 3 frames to test what happens when the memory is low, it works and prints out a poor result, however the variable global part does not print in a timely manner.

4. Review of Results and Development Observations

- During development my biggest issue was the fault times being slightly off for both local and global, this took a lot of time to fix as I was focused on the fault times, forgetting about the overall turnaround time. After considering these I realised that my turnaround was 1-3 faster than the main result, and using some AI's help I figured out this was a result of delay times not being added.
- As I noted in my testing methodology section I choose to use the same frames and quantum trough out development this was a terrible idea, after fixing by timing issue I noticed that the variable global method for sample 2 wasn't lining up everything else was fine though, so after hours of debugging and rewriting turns out I had forgotten to comprehend that due the size it needs more frames to achieve the times on the output sheet.